AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 5 · STUDENT LESSON

# Replication and reproduction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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## Before you start

Recall Methods results and limitations. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Explain the mechanism and terminology of replication and reproduction. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of **replication** and reproduction.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **replication** and reproduction

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Methods results and limitations. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

### This week's end goal

Explain **replication** and reproduction using the records and boundaries of this fictional investigation.

## Notice the evidence

> Investigate the supplied Replication and reproduction case. Separate observations, given facts and assumptions before choosing an explanation.

Reproducing a computation uses recorded artifacts; **replication** collects new **evidence** under a comparable question. Naming conventions vary, so define the terms in the project.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

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STUDENT LESSON · 3 OF 18

## See how it works

> Follow the mechanism for Replication and reproduction. Connect each labelled input, deciding step and result.

Here **replay** means same artifacts/procedure; new-data **replication** means newly collected eligible cases under the same plan. Preserve environment, data versions and raw results.

### Replication and reproduction — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. | Here replay means same artifacts/procedure; new-data replication means newly collected eligible cases under the same plan. Preserve environment, data versions and raw results. | This is a successful replay on eight distinct cases, not sixteen new cases. It checks computation consistency. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

> Compare the supplied cases for Replication and reproduction. Identify the detail that changes the conclusion.

### Compare a familiar case with a changed case

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

### Change one thing in the pictured case

#### Change one thing

Change: The rerun of the original records **output**s 2/4.

#### Predicted result

**Prediction**: Investigate code, version, **input** or scoring differences.

#### Why the result follows

Reason: The same fixed records should give the same deterministic count.

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STUDENT LESSON · 5 OF 18

## Words that unlock the idea

Use the vocabulary of Replication and reproduction to explain a concrete decision, rather than only memorising definitions.

| Word | Meaning |
| --- | --- |
| Replay | Repeating a computation with the recorded artifacts |
| Replication | A new evidence collection under a comparable declared question |
| Distinct unit | A separately eligible observation rather than a repeated copy |

**Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

## Follow a worked example

> Follow the worked trace for Replication and reproduction, predict a step, then compare with its explanation.

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases.

This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

### Worked explanation · Replication and reproduction

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Replication and reproduction

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

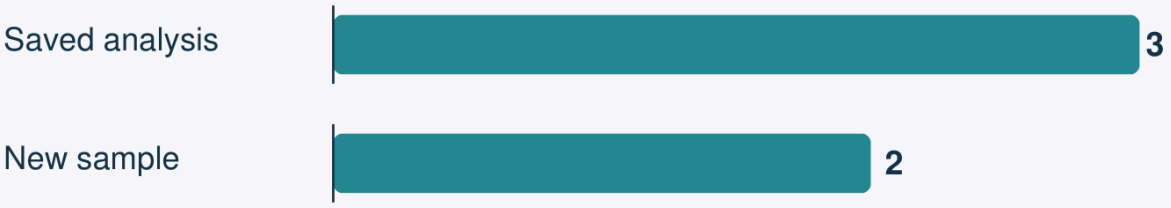
### The facts we will use

> **Supplied facts**
>
> Original recorded labels: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

AI ACADEMY / AI-7/5

# Different four-record datasets

correct cases out of 4; samples differ




Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/5

# Replication and reproduction

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Reproduce the analysis

Reusing the supplied four records gives 3/4=75%.

### 2. Repeat the protocol

The new four-record set gives 2/4=50%.

### 3. Compare purposes

The first checks computational reproduction; the second probes repeatability with new observations.

**CHANGE ONE THING**

The rerun of the original records outputs 2/4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Reproduce the analysis | Reusing the supplied four records gives 3/4=75%. |
| 2. Repeat the protocol | The new four-record set gives 2/4=50%. |
| 3. Compare purposes | The first checks computational reproduction; the second probes repeatability with new observations. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

### What this example does not establish

#### Keep the stated limit

Four-case samples are illustrative; neither result alone establishes broad reliability.

### Keep the stated limit

Read the labelled picture: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

### Keep the stated limit

Change one thing: The rerun of the original records **output**s 2/4. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Replication and reproduction claim and explain a repair using the supplied evidence.

Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

### Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Locate the earliest unsupported step

A report omits this qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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STUDENT LESSON · 8 OF 18

## Practise together

Practise Replication and reproduction with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Replication** and reproduction". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
|---|---|---|
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for **Replication** and reproduction. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

## Run the investigation

Investigate Replication and reproduction. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

### Predict, investigate and preserve the evidence

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |

| Record | Your evidence |
| --- | --- |
| Actual result, or “not run” | |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Replication and reproduction. Explain why each step is justified.

**Explain → apply → challenge**

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

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STUDENT LESSON · 11 OF 18

## Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **replication** and reproduction **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **replication** and reproduction.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

### Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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STUDENT LESSON · 12 OF 18

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

# Study a complete case

Revisit the complete Replication and reproduction case for supported practice. Repeated examples do not become fresh assessments.

## Complete evidence record

### Replication and reproduction — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. | Here replay means same artifacts/procedure; new-data replication means newly collected eligible cases under the same plan. Preserve environment, data versions and raw results. | This is a successful replay on eight distinct cases, not sixteen new cases. It checks computation consistency. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

**My complete trace of the worked case**

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STUDENT LESSON · 14 OF 18

## Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Apply a new case independently

Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

> Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

> Use this map to navigate Replication and reproduction, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of replication and reproduction | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Replication and reproduction without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **replication** and reproduction

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? The first checks computational reproduction; the second probes repeatability with new **observation**s.

1. Compare purposes

2. Reproduce the analysis

3. Repeat the protocol

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare purposes The first checks computational reproduction; the second probes repeatability with new **observation**s.

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The rerun of the original records **output**s 2/4. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Investigate code, version, **input** or scoring differences.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Investigate code, version, **input** or scoring differences. The same fixed records should give the same deterministic count.

**Repair a missing building block**

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner’s improvement.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

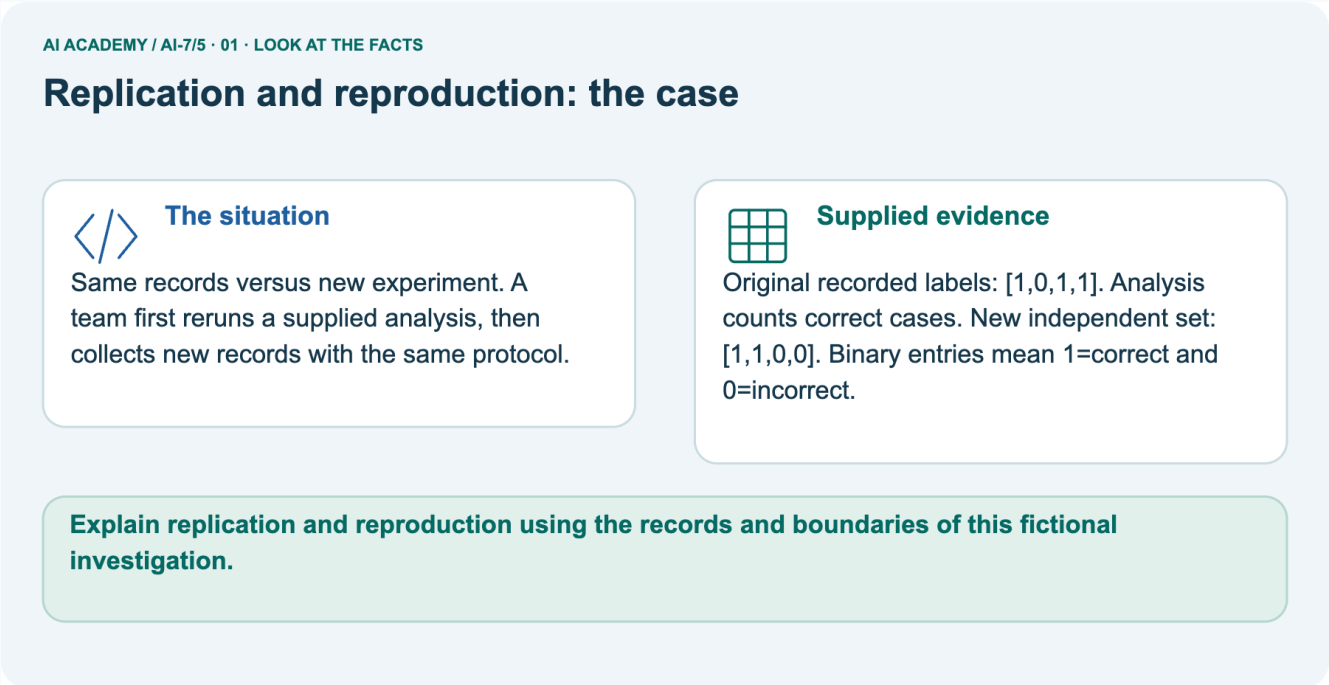
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STUDENT LESSON · 18 OF 18

# Visual concept journey

> Explain each Replication and reproduction graphic using its labels, deciding evidence and relevant limitation.

### 01. Replication and reproduction: the case



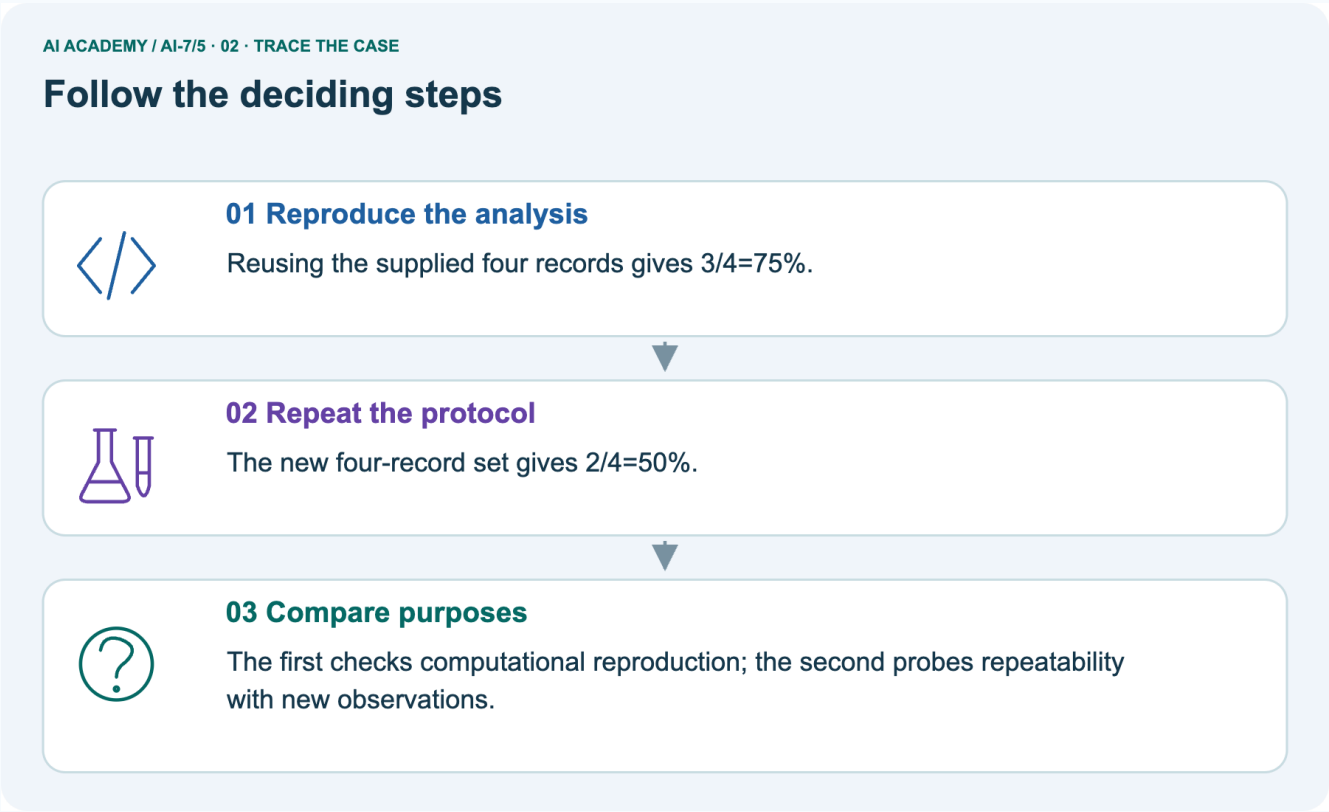

Follow the labelled picture with the matching explanation. Open the vector diagram

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol. Supplied facts: Original recorded labels: [1,0,1,1]. Analysis counts correct cases.

New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

> Point to the given fact that matters most. Explain why it matters.

### 02. Follow the deciding steps

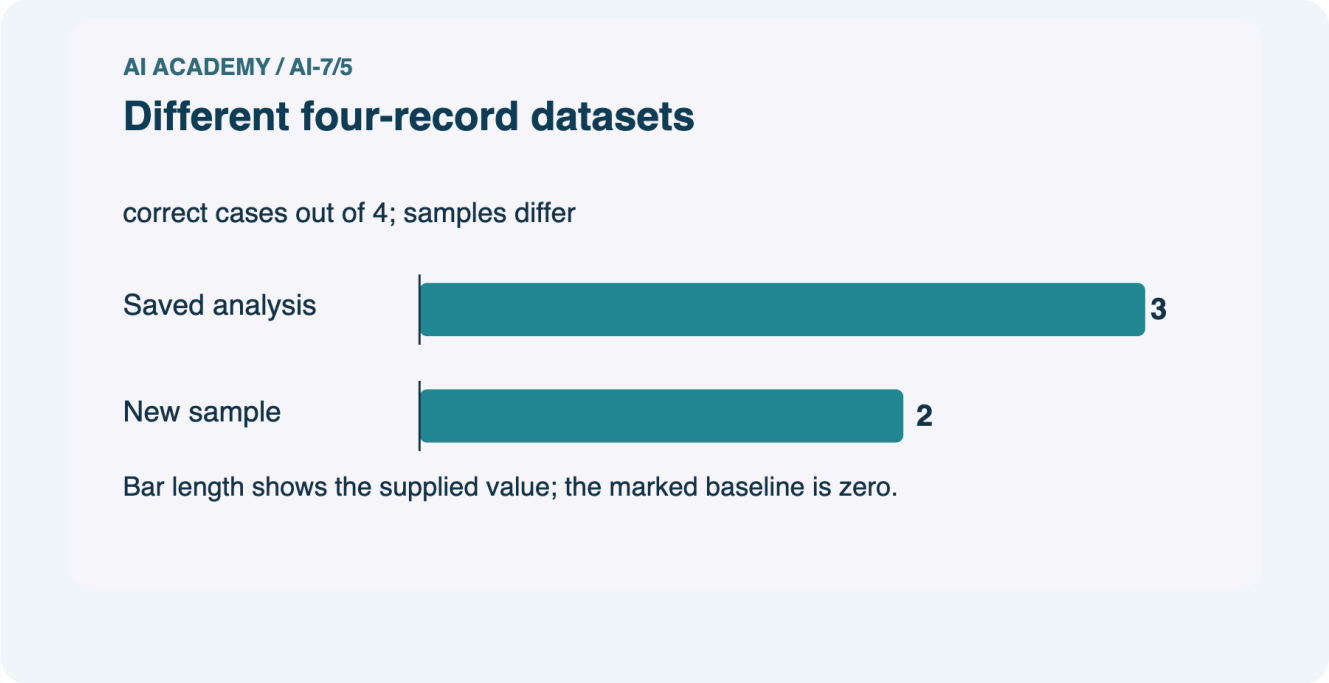



Follow the labelled picture with the matching explanation. Open the vector diagram

Reproduce the analysis: Reusing the supplied four records gives 3/4=75%. → Repeat the protocol: The new four-record set gives 2/4=50%. → Compare purposes: The first checks computational reproduction; the second probes repeatability with new **observation**s.

Point to each numbered stage and explain which recorded **input** it uses.

### 03. Different four-record datasets

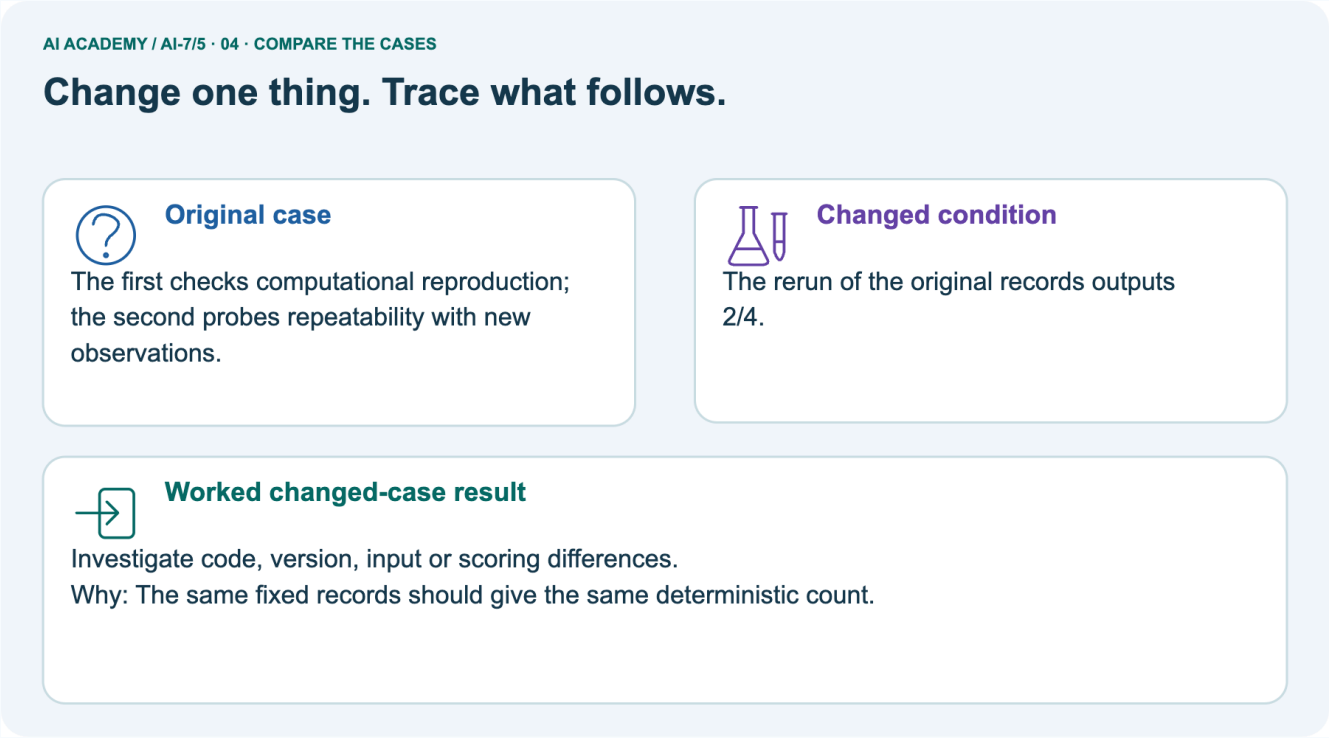



Follow the labelled picture with the matching explanation. Open the vector diagram

Different four-record datasets. Units: correct cases out of 4; samples differ. Saved analysis: 3. New sample: 2.

Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



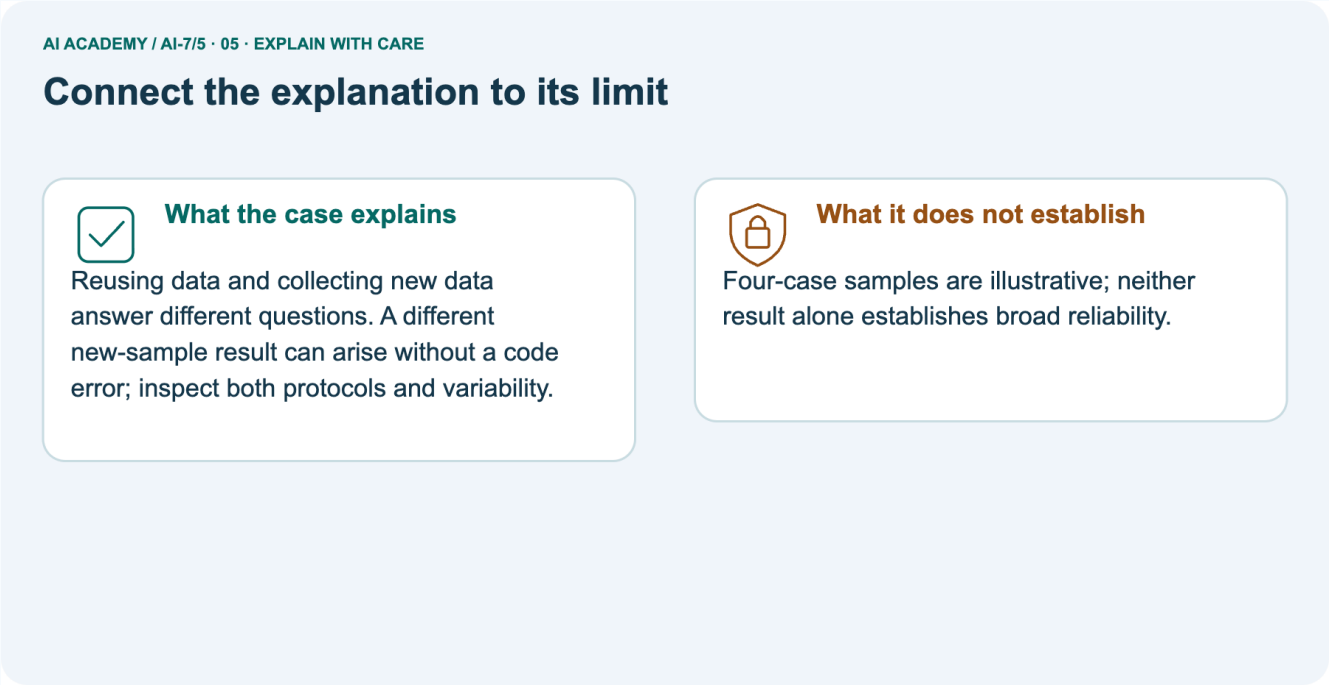

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The first checks computational reproduction; the second probes repeatability with new **observation**s. Changed condition: The rerun of the original records **output**s 2/4.

Expected result: Investigate code, version, **input** or scoring differences. Reason: The same fixed records should give the same deterministic count.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



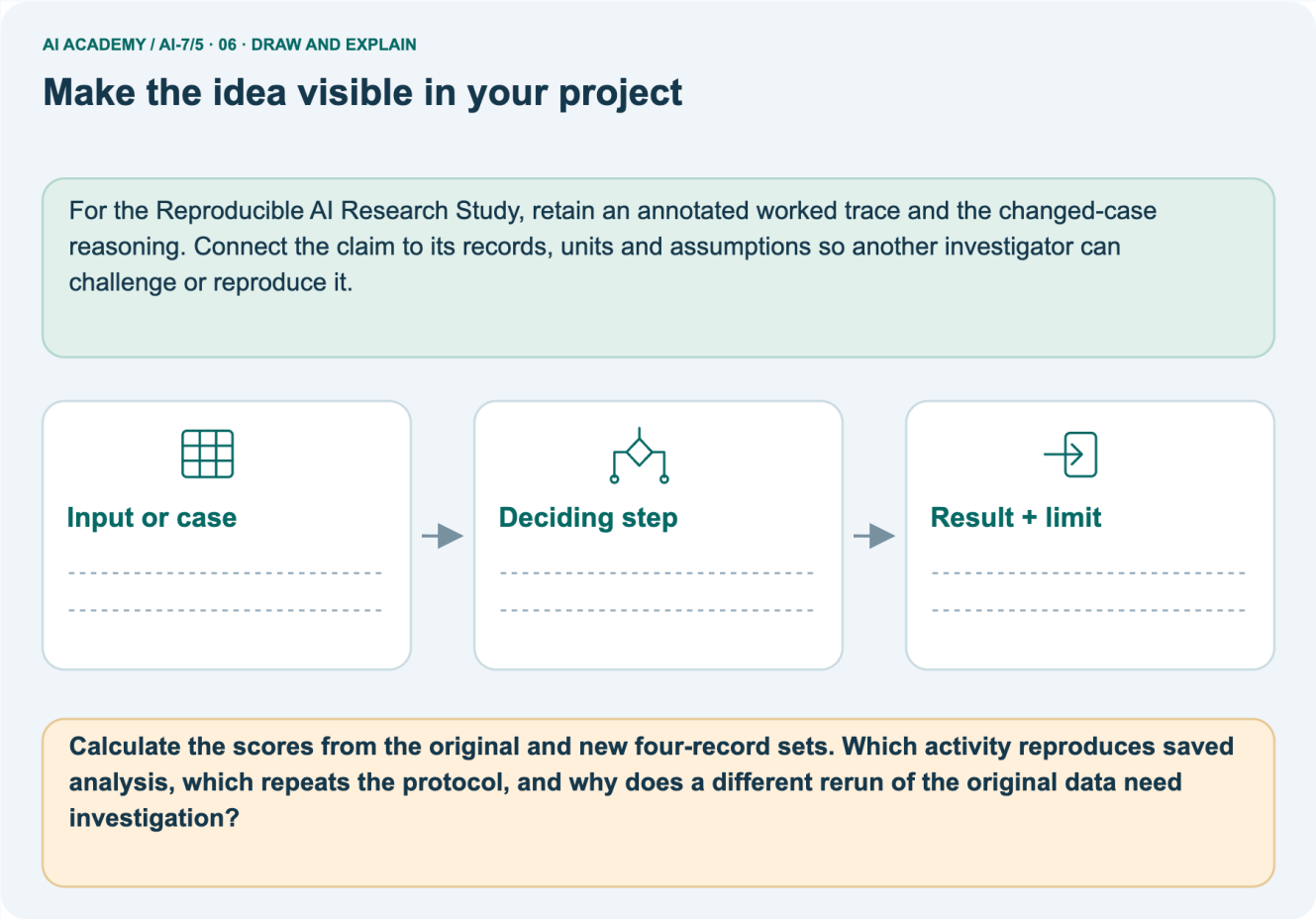

Follow the labelled picture with the matching explanation. Open the vector diagram

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Limit: Four-case samples are illustrative; neither result alone establishes broad reliability.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.